visualization of (a portion of) the 2-D cartesian plane, where the origin is at the center of the square.

We will start with the “easy” methods first: search and insert works almost exactly as in basic *BST*s; we will still describe them and provide their pseudocode, but if you are already familiar with binary search trees, feel free to skim through the next few sub-sections.

Listing 9.1 The KdTree class

```
class KdNode
  #type tuple(k)
  point
  #type KdNode
  left
  #type KdNode
  right
  #type integer
  Level
  function KdNode(point, left, right, level)

class KdTree
```

```
  #type KdNode
  root
```

```
  #type integer
  k
  function KdTree(points=[])
```

But before that, we need to define a model for the k-d tree and its nodes: check listing 9.1.